

SOGNEFJELL, Norway

WMO: 013660

Lat: 61.564N

Lon: 7.998E

Elev: 4636

StdP: 12.40

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.7	(c) -3.5	(d) -14.1	(e) 3.0	(f) -4.0	(g) -10.1	(h) 3.8	(i) -0.1	(j) 37.1	(k) 19.5	(l) 34.0	(m) 16.6	(n) 2.4	(o) 70	(p) 0.899

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.5	(c) 60.9	(d) 50.6	(e) 57.6	(f) 49.1	(g) 54.7	(h) 47.9	(i) 52.2	(j) 58.1	(k) 50.7	(l) 56.0	(m) 48.9	(n) 53.7	(o) 7.4	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
49.4	62.0	54.1	47.7	58.2	52.8	46.0	54.5	50.9	23.2	58.6	22.2	56.3	21.1	53.8	57.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
30.9	27.0	23.4	-11.1	62.7	5.4	2.9	-15.0	64.8	-18.2	66.5	-21.2	68.2	-25.1	70.3
			-11.8	53.8	5.3	2.8	-15.5	55.8	-18.6	57.4	-21.6	59.0	-25.4	61.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.4	14.2	15.5	17.5	24.8	32.5	39.9	46.1	44.6	38.3	29.0	21.0	16.7	
	DBStd	13.36	8.22	8.23	8.07	6.93	7.24	5.40	6.35	5.26	5.81	7.20	8.23	8.67	
	HDD50	7934	1111	967	1007	755	543	306	157	182	353	652	870	1031	
	HDD65	13355	1576	1387	1472	1205	1008	754	585	633	802	1117	1320	1496	
	CDD50	53	0	0	0	0	0	1	37	14	1	0	0	0	
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	9.7	11.0	10.1	11.1	9.7	8.4	7.7	7.8	8.3	9.9	10.0	10.2	11.6	
Precipitation	PrecAvg	31.3	3.5	2.7	2.6	1.4	1.5	1.9	2.5	2.7	2.8	3.1	3.2	3.6	
	PrecMax	43.3	9.1	6.3	11.1	2.5	3.6	4.5	4.3	4.3	4.7	7.6	6.5	7.5	
	PrecMin	19.4	0.2	0.0	0.1	0.5	0.3	0.4	1.0	0.5	0.8	1.1	0.4	0.6	
	PrecStd	5.2	2.2	1.7	2.1	0.6	0.7	1.0	0.9	0.9	1.1	1.5	1.8	1.9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.9	35.1	37.9	45.3	56.3	59.2	66.9	62.2	56.3	49.3	40.4	35.7	
		MCWB	30.9	32.1	31.2	37.0	45.7	46.5	53.8	50.2	47.7	44.8	35.5	35.3	
	2%	DB	28.9	31.6	32.9	40.8	50.4	55.9	63.1	58.1	52.2	43.3	36.7	32.3	
		MCWB	28.4	30.1	30.4	34.7	42.5	45.5	52.4	49.7	46.3	39.6	34.5	32.0	
	5%	DB	26.9	28.9	30.6	38.1	46.5	51.8	59.9	54.9	49.4	40.8	34.1	30.5	
		MCWB	26.2	27.7	29.1	33.8	40.1	43.7	50.7	48.9	45.4	37.7	32.5	30.1	
	10%	DB	25.1	26.4	28.2	35.4	43.0	48.8	56.5	52.6	46.8	38.1	31.3	27.9	
		MCWB	24.4	25.3	26.8	31.8	38.4	42.2	49.4	47.8	43.8	35.9	30.3	27.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.4	33.4	33.7	38.7	46.8	48.6	55.3	52.6	50.5	45.8	37.8	35.5
			MCDB	31.6	35.0	35.2	42.1	54.7	55.1	64.3	57.3	52.8	48.5	39.5	35.6
2%		WB	28.6	30.6	31.1	36.0	43.5	46.7	53.4	51.2	47.6	40.5	35.3	32.2	
		MCDB	28.7	31.2	32.2	39.1	48.8	54.1	60.4	56.0	50.6	42.5	36.3	32.4	
5%		WB	26.5	27.9	29.4	33.7	41.0	44.7	51.8	49.8	45.7	38.3	33.0	30.4	
		MCDB	26.8	28.7	30.7	36.9	45.5	50.8	58.1	54.3	48.7	40.0	34.0	30.6	
10%		WB	24.6	25.6	27.0	31.7	39.0	43.0	50.2	48.1	44.0	36.1	30.4	27.6	
		MCDB	25.1	26.3	28.3	34.6	42.7	47.7	55.9	52.3	46.8	37.7	31.5	28.0	
Mean Daily Temperature Range	5% DB	MDBR	8.9	9.8	11.0	11.8	10.4	10.6	9.5	7.9	7.6	7.9	8.8	9.3	
		MCDBR	8.2	10.5	11.7	14.7	15.0	16.2	14.4	11.0	11.5	10.5	9.2	9.8	
		MCWBR	8.1	9.9	9.8	10.8	8.9	9.6	7.2	5.7	7.1	8.2	8.2	10.0	
	5% WB	MCDBR	7.7	9.7	10.2	12.2	13.4	14.3	12.8	9.8	9.8	9.3	8.9	9.8	
		MCWBR	7.8	9.5	9.0	8.9	8.4	9.2	7.0	5.9	7.1	7.9	8.2	10.1	
Clear-Sky Solar Irradiance	taub	0.182	0.225	0.254	0.282	0.289	0.291	0.301	0.294	0.268	0.246	0.188	0.150		
	taud	2.110	2.273	2.257	2.245	2.330	2.484	2.491	2.505	2.561	2.429	2.125	1.944		
	Ebn at Noon	188	251	279	290	297	298	292	285	277	241	179	140		
	Edh at Noon	10	19	28	35	35	30	30	27	21	16	10	7		
All-Sky Solar Radiation	RadAvg	77	263	689	1253	1559	1575	1473	1132	711	326	104	46		
	RadStd	15	58	108	165	187	160	161	101	82	46	16	5		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (47 neighbors)	+0.72	+1.35	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page